CSV

View online: https://www.construct.net/en/make-games/manuals/construct-3/plugin-reference/csv

The **CSV** plugin supports reading and writing data in comma-separated values (CSV) format, as well as other similar formats that use a different delimiter such as tab-separated values (TSV).

Delimiter-based formats like CSV and TSV are simple and easy to use, but are limited in their ability to represent more complex data. For more advanced uses consider using JSON instead.

Scripting

When using JavaScript or TypeScript coding, the features of this object can be accessed via the ICSVObjectType script interface.

Using CSV files

CSV and TSV files can be added as project files in the Project Bar and edited directly in Construct. As with other data formats, these can then be loaded by requesting them with the AJAX object.

Using tab delimiters

Due to the use of the tab key as a shortcut in the editor, it can be difficult to type a tab in the Parameters Dialog for the delimiter parameter if you want to use tab-separated values. To make this easier the CSV plugin provides a *TabCharacter* expression which just returns a string with a single tab character in it, so it can be conveniently used as a delimiter parameter.

Using array for storage

The CSV plugin merely parses and generates CSV. It uses an Array object to store the actual data, as CSV can conveniently be represented as a two-dimensional array, and it allows using the full features of the Array object to read and manipulate the data.

CSV conditions

On parse error

Triggered when using the *Parse* action with invalid data, such as a quoted item that is missing its end quote. If a parse error occurs then no data will be read from the file and the specified Array object will be unmodified.

On parse success

Triggered when using the *Parse* action with valid data that is successfully parsed.

CSV actions

Generate CSV

Generate a string of CSV data using data from a specified two-dimensional Array object. A custom delimiter can be set for other delimiter-separated formats like tab-separated values. After this action the resulting string is available via the *GeneratedCSV* expression.

Parse CSV

Read a string of CSV data in to a specified Array object. The resulting data will be stored as a two-dimensional array. A custom delimiter can be set for other delimiter-separated formats like tab-separated values. If any of the rows are different lengths, the width of the array will be the maximum row length. If the data is invalid for any reason, *On parse error* will be triggered; otherwise if valid *On parse success* will be triggered. The *Data type* parameter determines whether values are read as strings or numbers. Its possible values are:

- **Auto:** automatically determine whether values are strings or numbers, based on whether the string looks like a number. For example "hello" will be stored as a string, but "100" will be stored as a number.
- **String:** treat every value as a string. Note this means a number like "100" will be stored as a string with that sequence of characters rather than an actual number type.
- **Number:** treat every value as a number. Note this means strings like "hello", which are not valid numbers, will result in the special "Not A Number" (NaN) value.

CSV expressions

GeneratedCSV

After the *Generate CSV* action, this expression returns a string of the generated CSV data.

TabCharacter

This expression merely returns a string with a single tab character in it. This can be useful to use as a delimiter parameter for tab-separated values (TSV), as a tab character can be difficult to type in the editor as it functions as a keyboard shortcut to move focus.